

Noise Robustness of Mutually Unbiased Bases under Compatibility Hypergraphs

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Abstract

We study the joint measurability of k noisy mutually unbiased bases (MUBs) in dimension d under a family of compatibility constraints parameterised by a hypergraph C . For $C = \{\{0, 1, \dots, m-1\}\}$ we recover the standard joint-measurability SDP of Designolle et al. [1], and we extend the analysis to intermediate hypergraphs interpolating between pairwise compatibility and full m -wise joint measurability, in the spirit of [2].

1 Introduction

2 Preliminaries

A positive operator-valued measure (POVM) on a d -dimensional Hilbert space $\mathbb{C}^d$ with outcome set $\{0, \dots, n-1\}$ is a family $M = \{M_a\}_{a=0}^{n-1}$ of Hermitian operators with

$$M_a \succeq 0, \quad \sum_{a=0}^{n-1} M_a = \mathbb{I}. \quad (1)$$

Given m POVMs $\{M^{(x)}\}_{x=0}^{m-1}$ with n outcomes each, they are *jointly measurable* iff there exists a joint POVM $\{G_\lambda\}_{\lambda \in [n]^m}$ such that

$$M_a^{(x)} = \sum_{\lambda: \lambda_x = a} G_\lambda \quad \forall x, a. \quad (2)$$

Definition 2.1 (Noisy MUB measurement). Let $\{|u_a^x\rangle\}_{a=0}^{d-1}$ for $x = 0, \dots, k-1$ be k mutually unbiased bases of $\mathbb{C}^d$. The η -noisy MUB measurement at setting x has elements

$$M_{a|x}^\eta = \eta |u_a^x\rangle\langle u_a^x| + (1-\eta) \frac{\mathbb{I}}{d}. \quad (3)$$

3 Compatibility hypergraphs

A *compatibility hypergraph* on m measurements is a collection $C = \{C_1, \dots, C_N\}$ with $C_i \subseteq \{0, \dots, m-1\}$. We say $\{M^{(x)}\}$ admits a C -decomposition iff

$$M_a^{(x)} = \sum_{i=1}^N M_a^{(x),i} \quad \forall x, a, \quad (4)$$

where each branch i contains a jointly measurable sub-family indexed by C_i and arbitrary POVMs for $x \notin C_i$. Equations (1) to (3) together with this definition yield the SDP studied in Section 4.

4 Numerical results

Table 1: Noise-robustness threshold $\eta^*(k, d)$ for k MUBs in dimension d . Tight values from Theorem 1 of [1] marked with † .

d	k	η^* (SDP)	note
2	2	0.7071	$1/\sqrt{2}^\dagger$
2	3	0.5774	$1/\sqrt{3}^\dagger$
3	4	0.5000	$1/\sqrt{4}^\dagger$
4	5	0.4472	$1/\sqrt{5}^\dagger$

5 Discussion

Acknowledgements

References

- [1] Sébastien Designolle, Paul Skrzypczyk, Florian Fröwis, and Nicolas Brunner. Quantifying measurement incompatibility of mutually unbiased bases. *Phys. Rev. Lett.*, 122(5):050402, 2019. doi: 10.1103/PhysRevLett.122.050402.
- [2] TODO: fill in authors. Todo: fill in title. *arXiv preprint*, 2019.